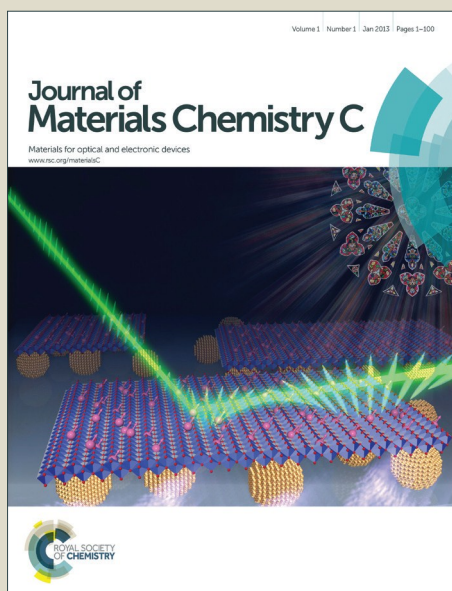


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Diaminomaleonitrile-Based Schiff Bases: Aggregation-Enhanced Emission, Red Fluorescence, Mechanochromism and Bioimaging Application

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Here we reported the synthesis of two diaminomaleonitrile-based Schiff bases with donor-acceptor structure and aggregation-enhanced emission feature. By changing the alkyl chain length in the donor unit, a red-emitting material with remarkable mechanochromic property and application in bioimaging was generated due to the J-aggregate formation in the solid state.

Luminescent organic materials with efficient emission in the solid state have received considerable attention due to their wide applications in optics, electronics, memory systems, biotechnology, etc.¹ Among them, red solid-state emitters are crucial for organic white lighting,² full color OLEDs³ and fluorescent imaging.⁴ Especially for bioimaging, red-emitting materials can circumvent the spectral overlap with the background autofluorescence, minimize photo-damage to living cells, and enable deep tissue penetration.⁵ General strategies to achieve red emission are either the endowment of π -conjugated molecules with planar conformation or the incorporation of electron donor (D) and acceptor (A) units into the molecular structure to induce intramolecular charge transfer (ICT) process.⁶ However, due to the strong π - π interactions or dipole-dipole interactions between the transition dipoles of D and A in traditional ICT systems, most of

these red-emitting materials suffer from the notorious aggregation-caused quenching (ACQ) effect: their bright emission in solution is partially or completely quenched upon aggregation.⁷ Aggregation-induced emission (AIE) is a photophysical phenomenon that was first proposed by our group and was exactly opposite to ACQ.⁸ Luminogens with AIE attributes (AIEgens) show little or no emission in dilute solutions but emit intensely when aggregated.⁹ Till now, numerous AIEgens have been developed, including tetraphenylethenes,¹⁰ siloles,⁸ fulvenes,¹¹ pyran derivatives,¹² conjugated polymers¹³ and others. These AIEgens have found a variety of high-tech applications.⁹ In particular, AIE-active dyes are promising materials used for bio-imaging as a result of their higher photostability and emission efficiency and better biocompatibility than traditional organic dyes and quantum dots.^{5a, 14} However, most of the AIEgens developed so far exhibit blue or green emission and are often prepared through complicated synthetic routes. Therefore, it is highly desirable to exploit AIEgens with efficient red emission in the solid state through facile synthetic procedures.

Schiff bases enjoy the advantages of rather straightforward synthesis and readily-modified structures. They are widely studied for their interesting biological and pharmacological activities. However, the exploration of their AIE potential is still in the early stage.¹⁵ Furthermore, to the best of our knowledge, apart from their applications in picric acid detection,¹⁶ data storage¹⁷ and chemo- or bio-probes for ions,^{15d, 18} few examples related to their use in cell imaging have been reported.¹⁹

Previously, we reported a diaminomaleonitrile (DMN)-functionalized Schiff base with diethylamino group (DEA, Scheme 1).²⁰ This molecule was found to exhibit both AIE and ICT features and emitted yellow light in the solid state. To widen its biological application, in this work, we modified its molecular structure to red-shift its emission color by changing the chain length in the electron-donating amino group. It is expected that such structural modification will alter the intermolecular distance to achieve emission shift. Surprisingly, results show that the molecule with dibutylamino group (DBA)

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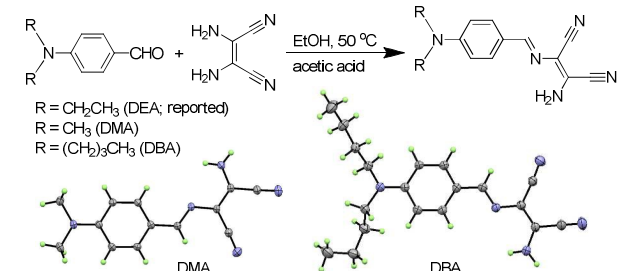
[†] T. Han and X. Gu contributed equally to this work.

Electronic Supplementary Information (ESI) available: Experimental details; single crystal data; fluorescence quantum yields; particle size analysis; TGA, TEM, UV, molecular amplitude plots of HOMO and LUMO, and molecular packing in single crystals of DMA and DBA; XRD data, MTT assay and photostability of DBA. CCDC 1491905 (for DMA) and 1491906 (for DBA). For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/x0xx00000x

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exhibit redder solid-state emission than that with a shorter dimethylamino group (DMA). Further study reveals that the red emission of DBA may origin from the formation of J-aggregates. Both DMA and DBA show aggregation-enhanced emission (AEE) properties. The red-emitting DBA also possesses remarkable mechanochromic property and can be used as a fluorescent visualizer for intracellular imaging.

DMA and DBA were synthesized by a one-step reaction under mild conditions in good yields (Scheme 1). This reaction exhibited high chemoselectivity²¹ and only one amino group of DMN was consumed even when an excess amount of aldehyde was added for the reaction. The structures of the products obtained were fully verified by IR, ¹H NMR, ¹³C NMR, ESI-MS spectroscopies and X-ray diffraction (XRD, Table S1 and S2). The detailed synthetic procedures and characterization data can be found in the Electronic Supporting Information.† Thermal gravimetric analysis showed that DMA and DBA possessed good thermal stability, losing 5% of their weight under nitrogen at 254 and 247 °C, respectively (Fig. S1).



Scheme 1 One-step synthetic route to DMN-functionalized Schiff bases and ORTEP drawings of single crystal structures of DMA and DBA with ellipsoids drawn at 50% probability.

The luminescent behaviors of DMA and DBA were then investigated and summarized in Fig. 1 and Table S3. The variation in the photoluminescence (PL) of DMA and DBA with the water fraction (f_w) in THF/water mixtures was similar. With an increase in water content, their PL intensity increases gradually and reaches their maximum value at f_w of 80% and 70%, respectively (Fig. 1). Analysis by dynamic light scattering suggests that nanoaggregates of DMA and DBA with sizes of about 904 and 1032 nm are formed, respectively (Fig. S2). In addition, the fluorescent quantum yield (Φ_f) of DMA and DBA increases from < 1% to 14.01% and 8.96%, respectively (Table S3). Evidently, DMA and DBA are AEE-active.²⁰ Similar to DEA, the space constraint in the aggregates may restrict the active intramolecular rotation of the phenyl rings and block the non-radiative relaxation of the excited states, thus enhancing the luminescence of the Schiff bases. It is noteworthy that further increment in f_w leads to a dramatic decrease in the emission intensity. To explain this, the morphologies of aggregates in solvent mixtures with f_w of 90% and 99% were examined by transmission electron microscopy. As shown in Fig. S3, DMA and DBA form microcrystals at f_w above 70% and 80%, respectively. Meanwhile, they show lower quantum yields in the crystalline state than in the pristine state (Table S3), presumably due to the stronger π - π interactions. Therefore, the emission drop observed at higher f_w could be attributed to

the π - π stacking interactions resulting from the crystallization process.²⁰

On the other hand, the emission maximum (λ_{em}) gradually red-shifts with increasing f_w (Fig. 1D), which may due to the activation of the ICT process induced by the enhancement of the environmental polarity upon water addition. To better understand this phenomenon, density functional theory calculations were performed based on the B3LYP/6-31G basis set. As shown in Fig. S4, the molecular orbital amplitude plots of HOMOs and LUMOs of DMA and DBA suggest that both of them show asymmetric electron density distributions. The HOMOs of DMA and DBA are dominated by the orbitals from the amino-substituted phenyl ring, while their LUMOs are contributed mainly by the electron-withdrawing maleonitrile unit. The calculation results clearly reveal the existence of strong charge transfer and hence ICT effect in these molecules.

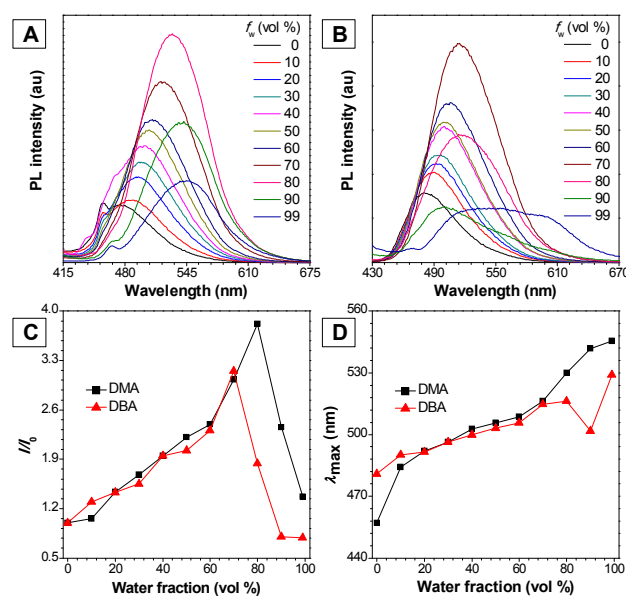


Fig. 1 (A and B) Photoluminescence (PL) spectra of (A) DMA and (B) DBA in THF/water mixtures with different water fractions (f_w). (C and D) Plots of (C) relative PL intensity (I/I_0) and (D) maximum emission wavelength (λ_{em}) versus the compositions of the THF/water mixtures of DMA and DBA. I_0 = PL intensity in pure THF solution. Concentration: 10 μ M; excitation wavelength: 400 nm.

The luminescent properties of DMA and DBA in the solid states are then further studied. As shown in Fig. 2, their emission can be tuned from yellow to red by modulating the alkyl chain length in the donor unit. Single crystals of DMA and DBA were grown and analyzed by XRD to understand the influence of molecular packing on their solid-state emission. Single-crystal XRD reveals that these molecules adopt a planar conformation (Fig. S5). The interplanar spacing in DEA crystals is around 3.358 and 3.382 Å.²⁰ Replacing the diethylamino group in DEA with dimethylamino group in DMA has shortened the interplanar spacing to 3.247 and 2.977 Å (Fig. S5A). Obviously, the π - π interactions between molecules becomes stronger when the chain length is shortened, resulting in emission red-shift from 563 to 578 nm.

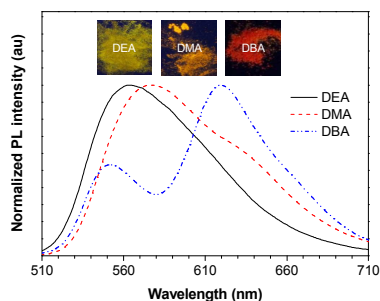


Fig. 2 Normalized PL spectra of powders of DMN-functionalized Schiff bases and their associated fluorescent photos taken under 365 nm UV irradiation.

Noteworthy, two emission peaks are observed in the PL spectrum of DBA in the solid state. Meanwhile, the DBA microcrystals absorb at redder region than their isolated species in THF solution (Fig. S6). Through careful inspection of the molecular packing in the crystal state (Fig. S5B), such behaviours may result from the J-aggregate formation. The peaks at 551 nm and 620 nm in the PL spectrum of DBA could be assigned to monomer and J-aggregate emission, respectively. As demonstrated in Fig. S5B, the electron-donating aromatic part stacks on the top of electron-withdrawing maleonitrile moiety at a distance of 3.141 Å. Two close molecules of DBA can form a dimeric unit through a head-to-tail fashion. Besides, as the transition dipole of the molecules rests along the long molecular axis, the adjacent dimeric units of DBA align in a staircase packing mode with a slippage angle of 20°. Such a large molecular displacement is due to the steric hindrance of the dibutyl amino group.²² This kind of arrangement belongs to a typical J-aggregation, thus explaining the observed band splitting in the PL spectrum and the bathochromic shift in both the absorption and emission spectra.

Mechanochromic AIE materials have gained tremendous attention owing to their practical applications in optical data storage, sensors, light-emitting devices, etc. A variety of mechanochromic AIEgens, such as phenanthroimidazole, benzothiazole, pyrenoimidazole, and tetraphenylethene derivatives, have been developed.²³ In this work, the emission response of DMA and DBA to external force was examined by grinding their powder using a spatula. Interestingly, DBA shows a mechanochromic feature with high contrast, whereas DMA does not (Fig. S7). When observed by naked eyes, the red powder of DBA is converted to a yellow one after gentle grinding. Its emission color is also altered from red ($\Phi_f = 8.96\%$) to yellow ($\Phi_f = 9.56\%$) under UV irradiation. This change only takes place at the ground region. A small video was taken to show the high sensitivity of this mechanochromic phenomenon (Video S1). Addition of a few drops of THF to the ground sample partially changes its color to the initial state (Fig. 3A and 3B). Analysis by PL spectroscopy demonstrates that the relative peak intensity rather than the location alters in the presence of external perturbation (Fig. 3C).

How to explain these phenomena? As suggested by single-crystal XRD (Fig. S5), the bulky dibutylamino group in DBA results in a loose packing, which provides room for the

molecules to adjust their conformations to right fashion by intramolecular rotation in response to external force. Once twisted conformations are formed, the π - π interactions will be damaged to some extent.²⁰ The consequent decomposition of the J-aggregates and generation of isolated monomers lead to the intensity rise of the emission peak at 551 nm and emission drop at 620 nm. This is also consistent with the slight increase in the Φ_f value after grinding. The addition and evaporation of THF solvent induce the recrystallization of the ground sample and promote the formation of J-aggregates again. As a consequence, the PL spectrum partially restore to the original pristine one. On the other hand, DMA exhibited more compact stacking and stronger intermolecular π - π interactions than DBA, thus making it difficult to adjust its molecular packing even when an external force was applied. Analysis by powder XRD (Fig. S8) suggests that the pristine DBA is well-ordered, as indicated by the clear and sharp diffraction peaks in its diffractogram. Upon grinding, the sample is amorphized and the spectrum now exhibits a few peaks with low intensity. After THF addition followed by subsequent evaporation, intense diffraction peaks emerge again, which indicates the regeneration of ordered packing in the sample. In short, the mechanochromic property of DBA is ascribed to the changes in molecular packing and morphology.²⁴

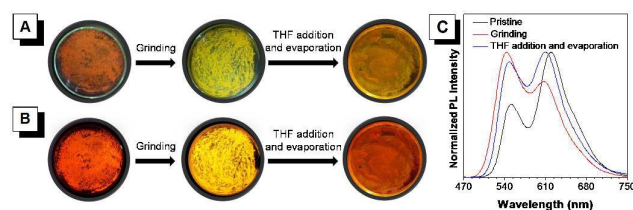


Fig. 3 (A and B) Switching the light emission of DBA by grinding and addition of THF solvent. The photos were taken under (A) daylight and (B) 365 nm UV light. (C) The relative PL spectra of the above processes.

The red emission and AEE property of DBA promote us to explore its application in cell imaging. As shown in Fig. 4 and Fig. S9, DBA can selectively stain the cytoplasm of living Hela cells. At an excitation wavelength of 540–580 nm, the image exhibit a bright red emission, which largely avoids the intracellular background autofluorescence. The cytotoxicity of DBA was assessed by using a 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyltetrazolium bromide (MTT) assay. As depicted in Fig. S10, no significant decrease in the cell viability was observed at a dye concentration of below 7.5 μM . When the concentration rises up to 10 μM , the cell viability is still high and reaches 88%, suggesting that DBA is biocompatible in the studied concentration range and shows low cytotoxicity to HeLa cells. The photostability of DBA was then examined (Fig. S11). After continuous excitation using Texas Red filter for 6 min, the emission of DBA remains almost unchanged, demonstrating its good photo-bleaching resistance. DBA is thus a good bioimaging agent with efficient red emission.

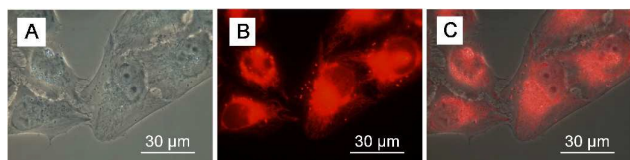


Fig. 4 (A) Bright-field and (B) fluorescent images of HeLa cells stained with 5 μM of DBA for 30 min. (C) Merged image of (A) and (B). Excitation wavelength: 540–580 nm.

In summary, two AEE-active molecules based on DMN-functionalized Schiff bases are designed and synthesized. Their solid-state emission can be tuned from yellow to red by controlling the formation of J-aggregate. DBA shows mechanochromic property and can be utilized as a fluorescent probe for bioimaging with low cytotoxicity and excellent photostability. The efficient red emission and one-step facile synthesis make DBA as a valuable AIEgen for bioimaging. Inspired by this, it is promising to generate more AIE-active and red-emissive Schiff bases for organelle-specific imaging in living cells through rational structure design. Further studies on the development of such materials are ongoing in our laboratory.

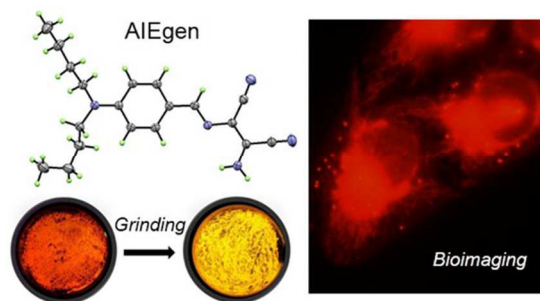
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Notes and references

- J. Mei, Y. N. Hong, J. W. Y. Lam, A. J. Qin, Y. H. Tang and B. Z. Tang, *Adv. Mater.*, 2014, **26**, 5429.
- W. Qin, J. W. Y. Lam, Z. Y. Yang, S. M. Chen, G. D. Liang, W. J. Zhao, H. S. Kwok and B. Z. Tang, *Chem. Commun.*, 2015, **51**, 7321.
- Q. S. Zhang, H. Kuwabara, W. J. Potscavage, S. P. Huang, Y. Hatae, T. Shibata and C. Adachi, *J. Am. Chem. Soc.*, 2014, **136**, 18070.
- (a) R. Weissleder and M. J. Pittet, *Nature*, 2008, **452**, 580; (b) C. F. A. Gomez-Duran, R. R. Hu, G. X. Feng, T. Z. Li, F. Bu, M. Arseneault, B. Liu, E. Pena-Cabrera and B. Z. Tang, *ACS Appl. Mater. Inter.*, 2015, **7**, 15168.
- (a) N. Zhao, S. J. Chen, Y. N. Hong and B. Z. Tang, *Chem. Commun.*, 2015, **51**, 13599; (b) X. M. Wu, X. R. Sun, Z. Q. Guo, J. B. Tang, Y. Q. Shen, T. D. James, H. Tian and W. H. Zhu, *J. Am. Chem. Soc.*, 2014, **136**, 3579; (c) A. Atilgan, E. T. Ecik, R. Guliyev, T. B. Uyar, S. Erbas-Cakmak and E. U. Akkaya, *Angew. Chem. Int. Ed.*, 2014, **53**, 10678.
- (a) M. Kim, D. R. Whang, J. Gierschner and S. Y. Park, *J. Mater. Chem. C*, 2015, **3**, 231; (b) M. Shimizu, R. Kaki, Y. Takeda, T. Hiyama, N. Nagai, H. Yamagishi and H. Furutani, *Angew. Chem. Int. Ed.*, 2012, **51**, 4095.
- (a) M. Shimizu and T. Hiyama, *Chem.-Asian J.*, 2010, **5**, 1516; (b) A. C. Grimsdale, K. L. Chan, R. E. Martin, P. G. Jokisz and A. B. Holmes, *Chem. Rev.*, 2009, **109**, 897; (c) J. Z. Liu, J. W. Y. Lam and B. Z. Tang, *Chem. Rev.*, 2009, **109**, 5799.
- J. D. Luo, Z. L. Xie, J. W. Y. Lam, L. Cheng, H. Y. Chen, C. F. Qiu, H. S. Kwok, X. W. Zhan, Y. Q. Liu, D. B. Zhu and B. Z. Tang, *Chem. Commun.*, 2001, 1740.
- J. Mei, N. L. C. Leung, R. T. K. Kwok, J. W. Y. Lam and B. Z. Tang, *Chem. Rev.*, 2015, **115**, 11718.
- Z. M. Wang, H. Nie, Z. Q. Yu, A. J. Qin, Z. J. Zhao and B. Z. Tang, *J. Mater. Chem. C*, 2015, **3**, 9103.
- (a) H. Tong, Y. Q. Dong, Y. N. Hong, M. Haussler, J. W. Y. Lam, H. H. Y. Sung, X. M. Yu, J. X. Sun, I. D. Williams, H. S. Kwok and B. Z. Tang, *J. Phys. Chem. C*, 2007, **111**, 2287; (b) X. G. Gu, J. J. Yao, G. X. Zhang, Y. L. Yan, C. Zhang, Q. Peng, Q. Liao, Y. S. Wu, Z. Z. Xu, Y. S. Zhao, H. B. Fu and D. Q. Zhang, *Adv. Funct. Mater.*, 2012, **22**, 4862. (c) Z. K. He, L. Q. Zhang, J. Mei, T. Zhang, J. W. Y. Lam, Z. G. Shuai, Y. Q. Dong and B. Z. Tang, *Chem. Mater.*, 2015, **27**, 6601.
- X. Y. Shi, H. Wang, T. Y. Han, X. Feng, B. Tong, J. B. Shi, J. G. Zhi and Y. P. Dong, *J. Mater. Chem.*, 2012, **22**, 19296.
- R. Hu, N. L. C. Leung and B. Z. Tang, *Chem. Soc. Rev.*, 2014, **43**, 4494.
- K. Wang, X. Y. Zhang, X. Q. Zhang, B. Yang, Z. Li, Q. S. Zhang, Z. F. Huang and Y. Wei, *Polym. Chem.*, 2015, **6**, 1360.
- (a) G. Liu, M. D. Yang, L. K. Wang, J. Zheng, H. P. Zhou, J. Y. Wu and Y. P. Tian, *J. Mater. Chem. C*, 2014, **2**, 2684; (b) A. Kathiravan, K. Sundaravel, M. Jaccob, G. Dhinakaran, A. Rameshkumar, D. A. Ananth and T. Sivasudha, *J. Phys. Chem. B*, 2014, **118**, 13573; (c) W. X. Tang, Y. Xiang and A. J. Tong, *J. Org. Chem.*, 2009, **74**, 2163; (d) W. Y. Fang, G. B. Zhang, J. Chen, L. Kong, L. M. Yang, H. Bi and J. X. Yang, *Sensor Actuat. B-Chem.*, 2016, **229**, 338; (e) J. H. Cheng, Y. X. Li, R. Sun, J. Y. Liu, F. Gou, X. G. Zhou, H. F. Xiang and J. Liu, *J. Mater. Chem. C*, 2015, **3**, 11099; (f) G. F. Li, W. Guan, S. Du, D. X. Zhu, G. G. Shan, X. J. Zhu, L. K. Yan, Z. M. Su, M. R. Bryce and A. P. Monkman, *Chem. Commun.*, 2015, **51**, 16924; (g) G. F. Li, X. Y. Ren, G. G. Shan, W. L. Che, D. X. Zhu, L. K. Yan, Z. M. Su and M. R. Bryce, *Chem. Commun.*, 2015, **51**, 13036.
- (a) V. Berau, C. Duhayon and J. P. Sutter, *Chem. Commun.*, 2014, **50**, 12061; (b) K. Acharyya and P. S. Mukherjee, *Chem. Commun.*, 2014, **50**, 15788.
- T. Y. Han, X. Feng, D. D. Chen and Y. P. Dong, *J. Mater. Chem. C*, 2015, **3**, 7446.
- (a) I. H. Hwang, Y. W. Choi, K. B. Kim, G. J. Park, J. J. Lee, L. Nguyen, I. Noh and C. Kim, *New J. Chem.*, 2016, **40**, 171; (b) J. Tang, Y. B. Cai, J. Jing and J. L. Zhang, *Chem. Sci.*, 2015, **6**, 2389.
- H. D. Xiao, K. Chen, D. D. Cui, N. N. Jiang, G. Yin, J. Wang and R. Y. Wang, *New J. Chem.*, 2014, **38**, 2386.
- T. Y. Han, Y. N. Hong, N. Xie, S. J. Chen, N. Zhao, E. G. Zhao, J. W. Y. Lam, H. H. Y. Sung, Y. P. Dong, B. Tong and B. Z. Tang, *J. Mater. Chem. C*, 2013, **1**, 7314.
- A. Rivera, J. Rios-Motta and F. Leon, *Molecules*, 2006, **11**, 858.
- X. Y. Qiu, S. H. Han, Y. F. Hu and B. W. Sun, *Chem.-Eur. J.*, 2015, **21**, 4126.
- (a) B. J. Xu, J. J. He, Y. X. Mu, Q. Z. Zhu, S. K. Wu, Y. F. Wang, Y. Zhang, C. J. Jin, C. C. Lo, Z. G. Chi, A. Lien, S. W. Liu and J. R. Xu, *Chem. Sci.*, 2015, **6**, 3236; (b) T. Jadhav, J. M. Choi, B. Dhokale, S. M. Mobin, J. Y. Lee and R. Misra, *J. Phys. Chem. C*, 2016, **120**, 18487; (c) R. Misra, T. Jadhav, B. Dhokale and S. M. Mobin, *Chem. Commun.*, 2014, **50**, 9076; (d) T. Jadhav, B. Dhokale, S. M. Mobin and R. Misra, *J. Mater. Chem. C*, 2015, **3**, 9981; (e) T. Jadhav, B. Dhokale, S. M. Mobin and R. Misra, *RSC Adv.*, 2015, **5**, 29878; (f) T. Jadhav, B. Dhokale and R. Misra, *J. Mater. Chem. C*, 2015, **3**, 9063.
- (a) Y. Q. Dong, J. W. Y. Lam and B. Z. Tang, *J. Phys. Chem. Lett.*, 2015, **6**, 3429; (b) Y. Sagara and T. Kato, *Nat. Chem.*, 2009, **1**, 605.

*Table of Contents Entry***Diaminomaleonitrile-Based Schiff Bases: Aggregation-Enhanced Emission, Red Fluorescence, Mechanochromism and Bioimaging Application**

Ting Han, Xinggui Gu, Jacky W. Y. Lam, Anakin C. S. Leung, Ryan T. K. Kwok, Tianyu Han, Bin Tong, Jianbing Shi, Yuping Dong,* and Ben Zhong Tang*



Red-emissive diaminomaleonitrile-based Schiff bases with aggregation-enhanced emission, mechanochromism, and application in bioimaging were designed and readily synthesized.